

The Robot

The espionage robot resembles an Armoured Personnel Carrier (APC). Its basic task is to go out looking for enemy troops and transmit what it finds in audio-visual format, using AM signals. The robot can operate in varying terrain—land, air and water—and has an arm that mimics the function of a human arm, a camera to see with, and a microphone to record sound.

On the top, the robot has rotor wings, propeller heads and radar, the Ariel Eyeball, and the robotic arm. The base of the robot has a mine detection and deactivation circuit, and mattracks similar to those in an APC.

The Military Suit

This consists of two control units on either arm—the one on the left controls motion and the one on the right controls the robotic arm. The suit also includes LCD goggles and a headphone-mic combination.

The waist belt carries the voice recognition system, batteries, and a signal receiver. The voice recognition system acts as a security device so that only the chosen soldier can control the robot.

The Base Camp Computer

This acts as a monitoring tool for the back-end forces and higher officials. It is connected to the robot and gets all the data. It needs at least Windows 98 with Turbo C installed. In case anything goes wrong with the soldier who controls the robot, then the Base Camp Computer can control the robot as well as transfer the control from one soldier to another.

Air, Land, and Sea

Shadab says, "It was tough for me to make the robot fly, initially. Since the entire body is quite heavy, battery consumption would be



The creation—his ultimate Soldier Gear

high if I had to make the robot fly. That's why I segregated the robot into an aerodynamically-flying saucer that sits atop the robot."

Another innovative feature added to the espionage robot was the ability to swim. "This wasn't as easy as it sounds. Dip an empty glass into water with the mouth of glass facing down and you will see that the inner wall of the glass stays dry. I applied the same rule for the robot", says Shadab. He modified the body of a vacuum cleaner to make the robot body, and used a large tank to test it. He added propellers (like those used in submarines) to give it thrust. It worked!

Next, he wanted the robot to climb, mainly trees. This would help the robot get higher up and improve its chances of spotting enemy troops. The two mattracks on either side have rubber and spiked grips are used for climbing. When it moves forward, it's the rubber that grips, and when it moves backwards, the spikes grip. So it goes up a tree backwards. The movement of the mattracks on either side can be individually controlled.

The Accolades

The Institute of Electronics and Electrical Engineers (IEEE) veterans Society of India awarded him a certificate and Rs 10,000 for his invention. He has already applied for a patent for his invention at the SIDBI Innovation and Incubation Centre, IIT Kanpur.

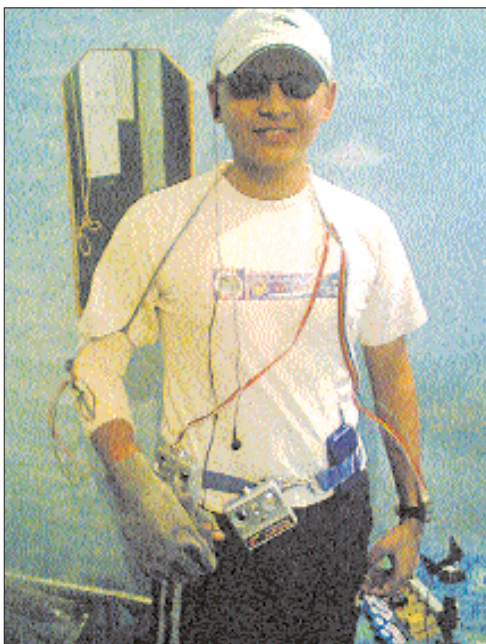
The Defence Research and Development Organisation (DRDO) has invited him to demonstrate his invention, but for that he wants to improve his robot using better-quality parts.

The new model will feature a few more functions such as a Global Positioning System and Shadab wants to add inter-connectivity between two or more Soldier Gears, to improve co-ordination and efficiency.

When asked about future plans, Shadab says, "I've finished my B. Tech in Electronics and Communications, but have no concrete plans to go commercial. At present, I seek funds to make the next Soldier Gear better."

Well done Shadab, we wish you success! ■

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The man in his military suit